

Sakhinala Sanjay Bhargav

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Summary

AI/ML Engineer specializing in Generative AI, agentic RAG systems, LLM evaluation, and reinforcement learning. Designed and built end-to-end AI systems using PyTorch, LangGraph, Hugging Face, CrewAI, and FastAPI, including a multilingual agentic RAG platform supporting 10 Indian languages with citation grounding, claim-level faithfulness evaluation, and offline evaluation pipelines. Developed a reinforcement learning framework that reduced electromagnetic simulation cost by 72% while outperforming five benchmark optimizers on CPU. Author of three IEEE publications and one Indian patent application.

Technical Skills

- **Languages:** Python, SQL
- **Machine Learning & RL:** PyTorch, TensorFlow, Scikit-Learn, LightGBM, Stable-Baselines3, Optuna, Deep Learning, Reinforcement Learning, Computer Vision, Neural Networks, OpenCV
- **NLP & Generative AI:** Retrieval-Augmented Generation (RAG), Large Language Models (LLM), Agentic AI, Multi-Agent Systems, LLM Evaluation & Observability, Fine-tuning, Prompt Engineering, Semantic Search, HuggingFace Transformers, LangChain, LangGraph, CrewAI, BGE-M3, Multilingual Embeddings, OCR
- **Backend & MLOps:** FastAPI, SQLAlchemy 2.0, REST APIs, WebSocket, Docker, Docker Compose, CI/CD, Nginx, PostgreSQL, pgvector, ChromaDB, FAISS, Redis, MLflow, Prometheus, Model Deployment, Google Cloud Platform, Vertex AI, Git, Linux
- **Data:** Pandas, NumPy, SciPy, Matplotlib, Streamlit

Projects

IndicRAG — Multilingual Agentic RAG System

Nov 2025 – Present

github.com/DNSdecoded/IndicRAG | [Blog](#) | [DeepWiki](#)

- Built a self-correcting research agent as a LangGraph StateGraph (planner → tool selector → executor → generator → reflexion evaluator) querying a local corpus, arXiv, Semantic Scholar/OpenAlex, and the web through 6 tools.
- Reached **Recall@5 0.917**, **citation grounding 0.938**, and **under 2% hallucinations** across [24] hand-labeled queries in 12 languages (English + 11 Indic), using hybrid BGE-M3 dense and BM25 retrieval fused with RRF (k=60), cross-encoder and ColBERT MaxSim reranking, and NLI claim-level faithfulness verification.
- Sustained **100+ commits through v2.4.0** over 9 months of solo development with PR-based review, semantic versioning, and documented rollback procedures; deployed with three-tier TTL/LRU caching, Prometheus metrics, and Nginx via Docker Compose.

Meridian UM — AI-Augmented Prior Authorization Platform

Jul 2026

github.com/DNSdecoded/meridian-um-platform | Synthetic data only

- Enforced citation grounding at the schema level so that **no LLM finding reaches a human reviewer unsourced** — findings must resolve to a document span or policy-clause ID, uncited findings are rejected, and a PostgreSQL CHECK constraint blocks any denial lacking recorded human attestation.
- Delivered a hole-free audit trail across an **11-state forward-only workflow** (step-skipping rejected, repeats idempotent) on async FastAPI and SQLAlchemy 2.0 over PostgreSQL 16 with pgvector policy-clause embeddings, OCR span indexing, role gating, rate limiting, and JWT verification.
- Made prompt changes measurable with an offline eval harness reporting accuracy, per-label precision/recall/F1, and citation validity over **~150 labeled synthetic cases**, backed by a provider-agnostic LLM adapter with failover and key rotation.

AgentOps — LLM Agent Observability & Evaluation Platform

Jul 2026

github.com/DNSdecoded/AgentOps | [Live demo](#)

- Built an agent tracing platform over a **12-table trace schema** exposed via REST and WebSocket, rendering planner → tool → retrieval → LLM executions as an interactive execution graph with per-model cost and per-stage latency rollups.
- Made quality regressions quantifiable through a dual-path evaluation engine — zero-cost token-overlap heuristics, swappable to an LLM judge — scoring faithfulness, retrieval relevance, and hallucination, plus a prompt A/B lab comparing versions on success rate, cost, and latency.

Reinforcement Learning for Antenna Design Optimization

Aug 2025 – Jan 2026

Presented, WAMS 2026 | sanjaysakhinala.pages.dev/blog/rl-antenna-design

- Cut electromagnetic simulation cost **72%** ($\sim 1,500 \rightarrow 420$ **CST Studio evaluations**) by training a Soft Actor-Critic agent over 2.68M steps against a blended surrogate ensemble ($2 \times \text{LightGBM} + \text{MLP} + \text{RidgeCV}$, < 5 ms inference) with uncertainty-aware reward shaping and an active-learning loop that re-simulates top candidates to limit model drift.
- Outperformed **all 5 benchmark optimizers** (GA, PSO, DQN, PPO, DDPG) at 28 GHz — -52.2 dB return loss, 8.93% FBW, 5.272 dBi gain, 80.6% radiation efficiency — as the only method combining surrogate acceleration, uncertainty quantification, and fabrication constraints.
- Demonstrated **GPU-free convergence** on a quad-core CPU workstation using curriculum learning ($32 \rightarrow 64 \rightarrow 128$ step episodes) with VecNormalize; mean FBW improved $8.1\% \rightarrow 9.0\%$ with policy variance below 0.01 pp.

Additional Projects

Jul 2025 – Jul 2026

- **AI Data Scientist** (Jul 2026) — Seven sequential CrewAI agents take a raw dataset to cleaned data, trained models, charts, and a written report; every agent is bound to deterministic Pandas/Scikit-Learn/SciPy tools so the LLM orchestrates while the tools compute. SQLite experiment store, LiteLLM provider abstraction, backoff retry on HTTP 429.
- **EuroSAT Land Use Classification** (Jul 2025) — Custom 3-layer CNN trained from scratch in PyTorch reaching **89% test accuracy** on 27,000 satellite images across 10 classes; Grad-CAM hooks on the final Conv2d layer verify the model learns spatial rather than texture features.
- **CipherChat** (Dec 2025) — End-to-end encrypted messenger implementing the Signal Protocol (X3DH key agreement and Double Ratchet) with AES-256-GCM, TLS transport, perfect forward secrecy, and TOFU key verification.

Publications & Patent

- **Uncertainty-Aware Reinforcement Learning System with Blended Surrogate Models for Electromagnetic Structure Optimization** — Presented, WAMS 2026, BVRIT Hyderabad. SAC with LightGBM/MLP/RidgeCV ensemble; 72% fewer simulations than a GA baseline.
- **Bandwidth Enhancement of Slotted Hexagonal Patch Antenna for 6G and Brain-Computer Interface** — ICMOCE 2025, IIT Bhubaneswar. 18.33 GHz bandwidth, 90.91% efficiency. DOI: [10.1109/ICMOCE64100.2025.11076991](https://doi.org/10.1109/ICMOCE64100.2025.11076991)
- **Bandwidth Optimization of Slotted Circular Patch Antenna for 6G and Brain-Computer Interface** — 16th ICCNT 2025, IIT Indore. 17.69 GHz bandwidth, 47.06 dB return loss. IEEE Xplore publication pending.
- **Patent application:** Design of Hexagonal Patch Antenna at 28 GHz — Indian application No. 202541014595 A, published Mar 2025.

Education

B.Tech, Electronics and Communication Engineering

May 2025

Bapatla Engineering College, Bapatla, Andhra Pradesh

Diploma, Electronics and Communication Engineering

Oct 2021

Aditya College of Engineering, Surampalem, Andhra Pradesh

Certifications & Training

- **Generative AI** — SkillHive Connect (Aug 2025 – Dec 2025). Fine-tuned LLaMA-derived models for domain-specific QA; built a voice-enabled medical assistant and a semantic search pipeline with sub-100 ms retrieval.
- **AI/ML** — Google for Developers (Jan 2025 – Mar 2025). End-to-end ML deployment pipelines on Vertex AI and real-time inference workflows integrating models into application backends.
- **Additional:** Google Data Analytics Professional Certificate; Google Cybersecurity Professional Certificate; Palo Alto Networks Cybersecurity; Altair Data Science Training; DeepLearning.AI AI For Everyone; NPTEL Cloud Computing; Google Cloud skill badges (TensorFlow on GCP, Image Classification, Object Detection, Vertex AI Pipelines).